

Visualization Examples

Example 1:

If we see in pic. The chosen values are:

$$w = -0.15, b = 800$$

→ Top-Left:

- At $x = 0$: $f(0) = -0.15(0) + 800 = 800$

so line starts at 800.

- Slope $= -0.15$ → Negative slope, so as x increases the line goes DOWN.

also bad fit for data as you can see.

→ Top-Right:

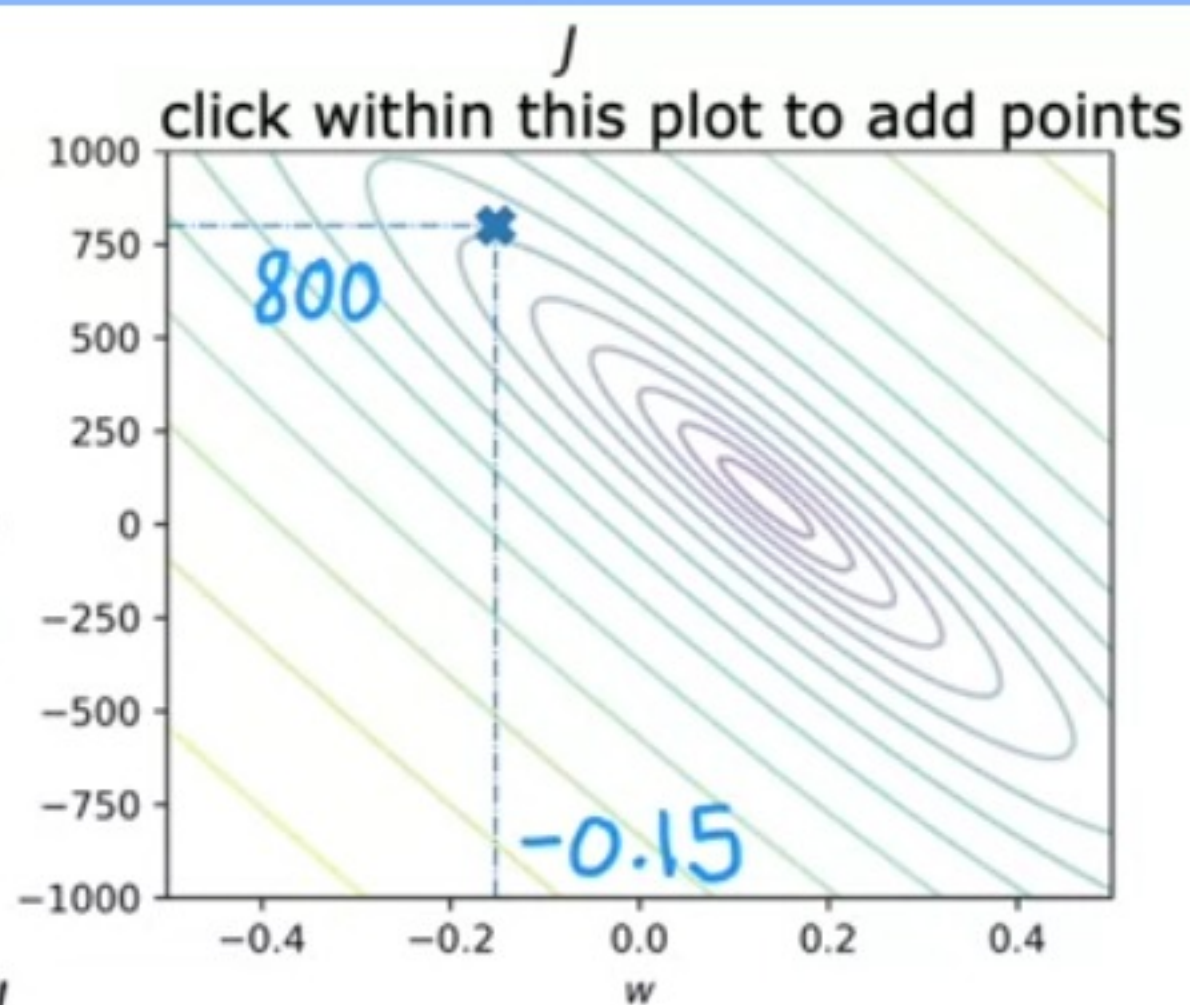
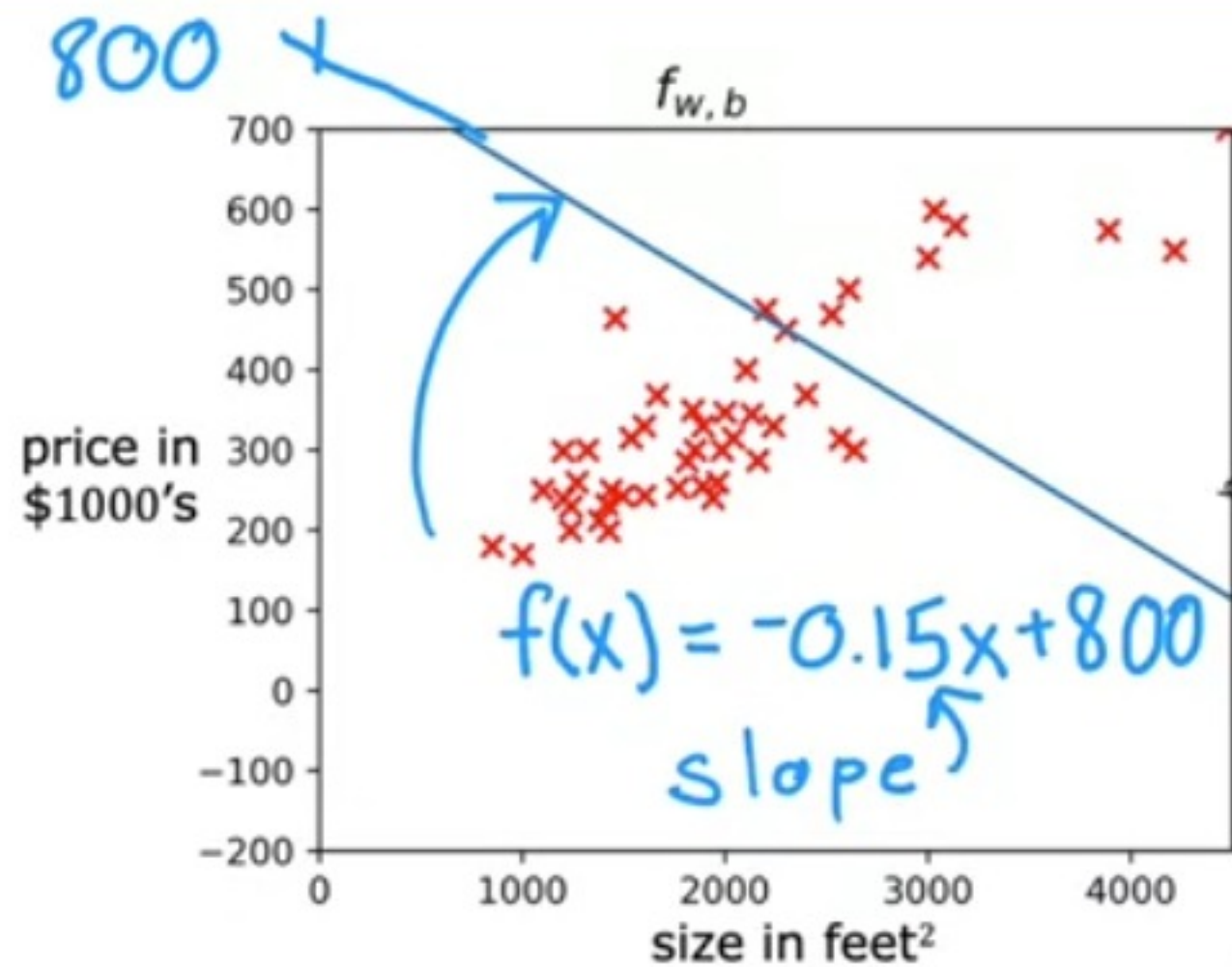
b-axis (vertical) → 800

w-axis (horizontal) → 0.15

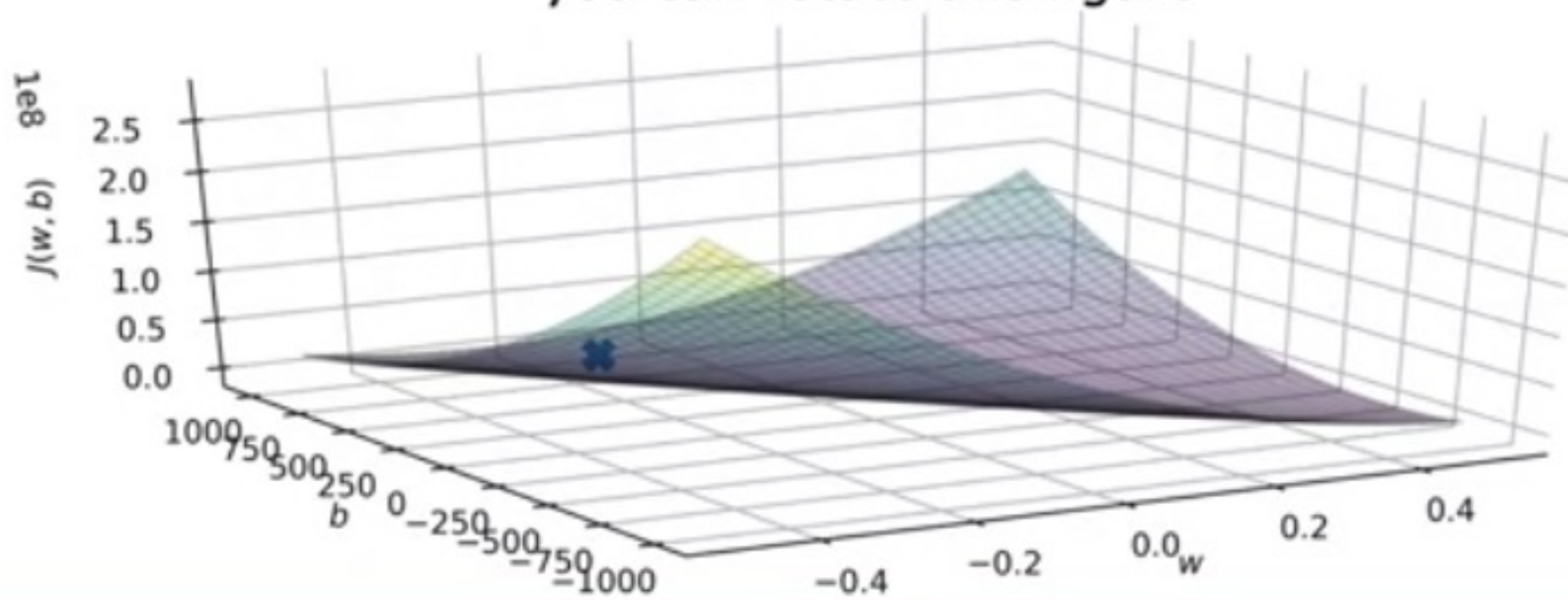
→ where these intersect - that's point as not near centre (bad fit).

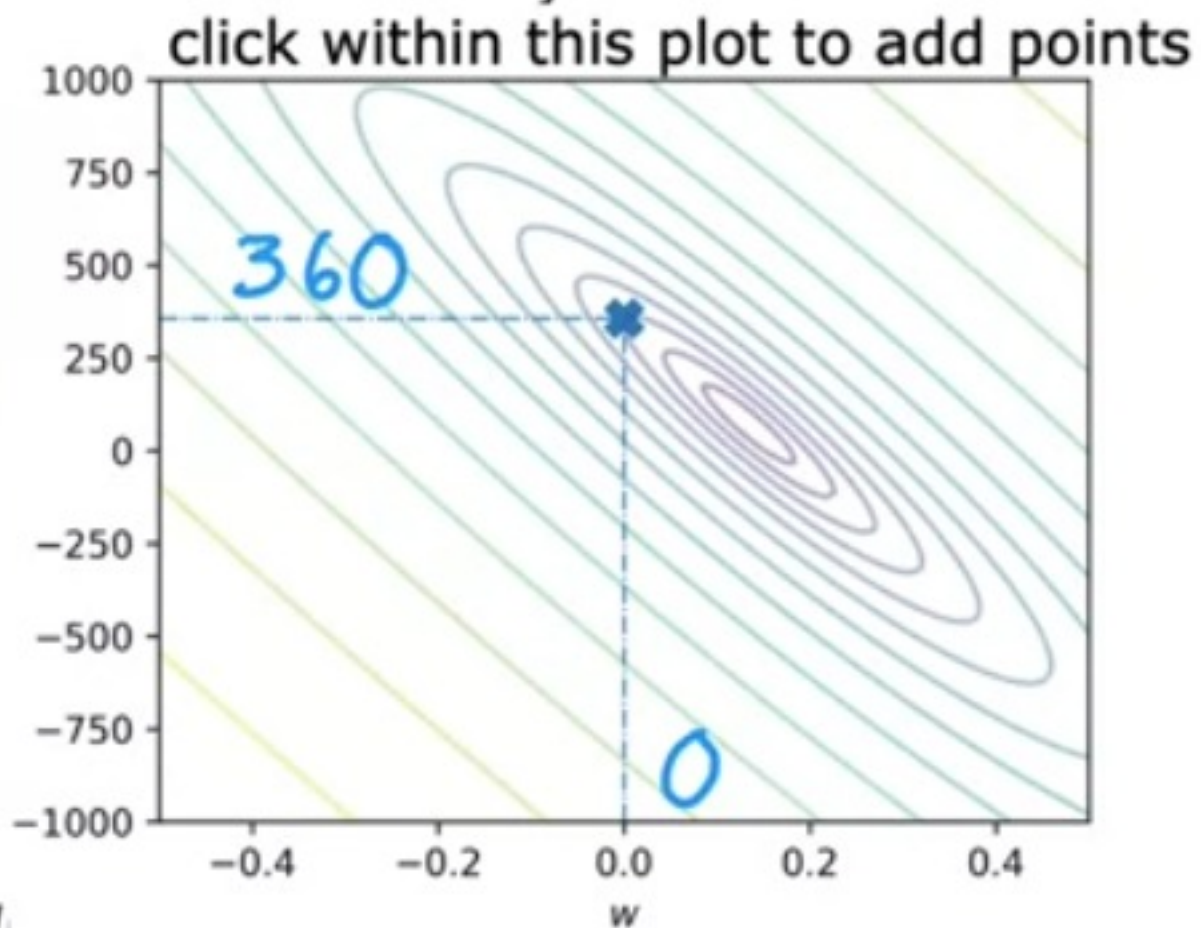
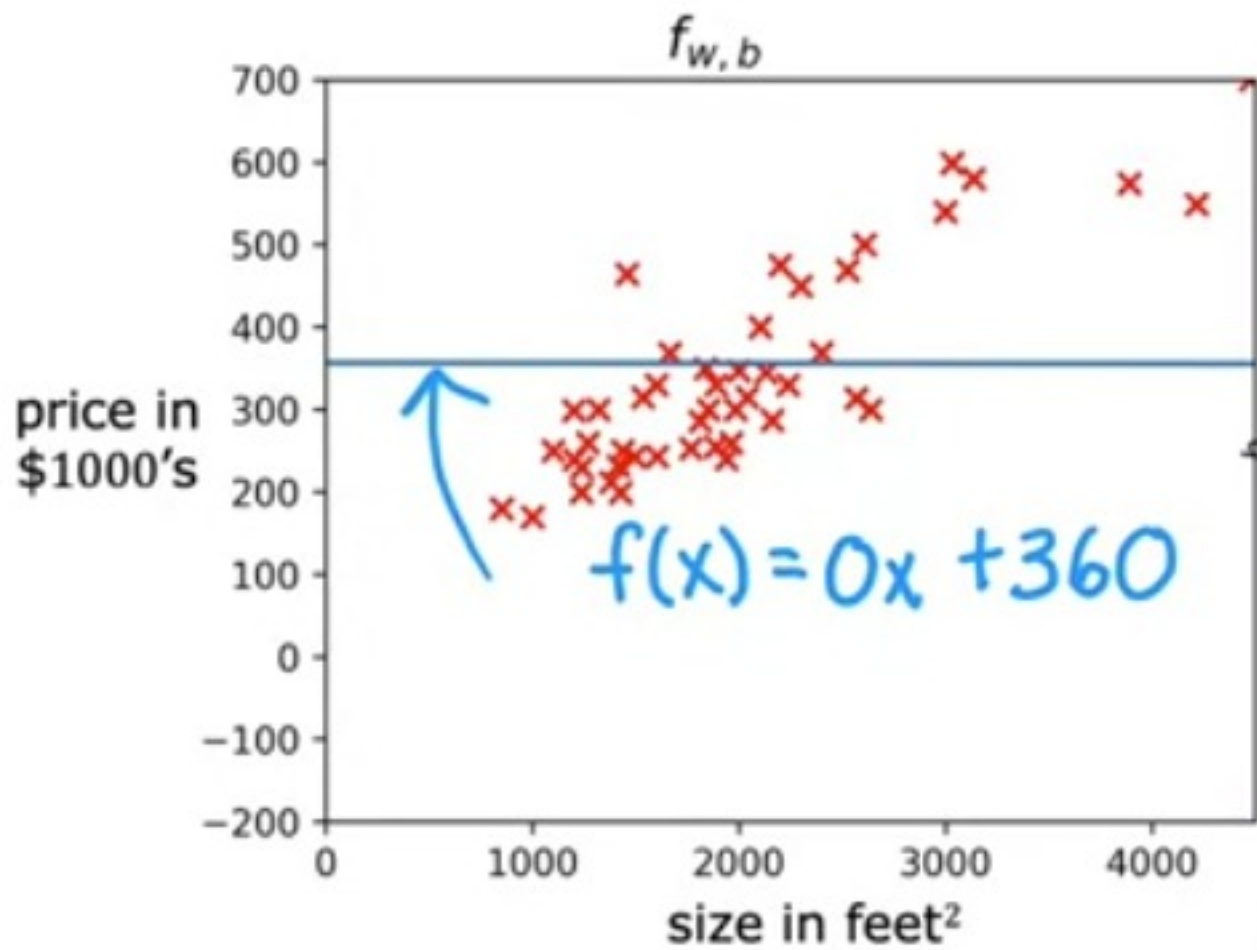
→ ~~Top-Left~~: Bottom:

same way but instead of contour, it's on 3D bowl shape, as it's sits high up not bottom (bad fit)

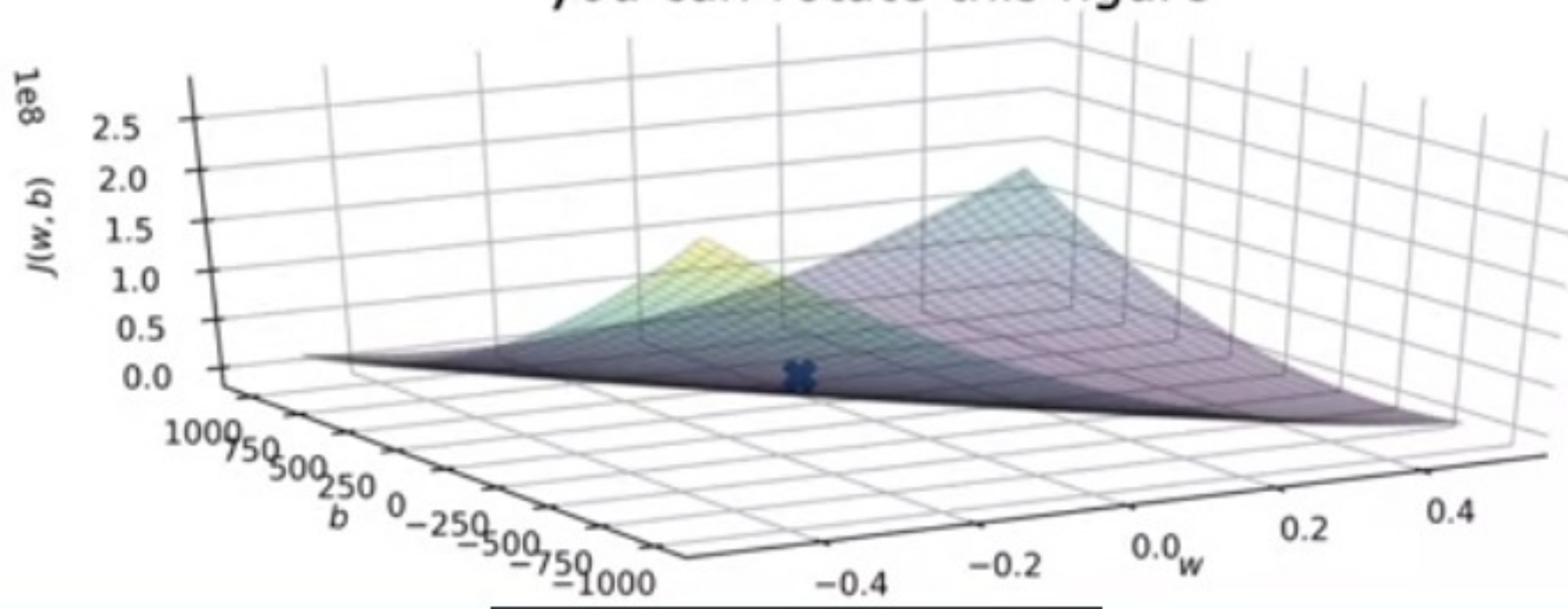


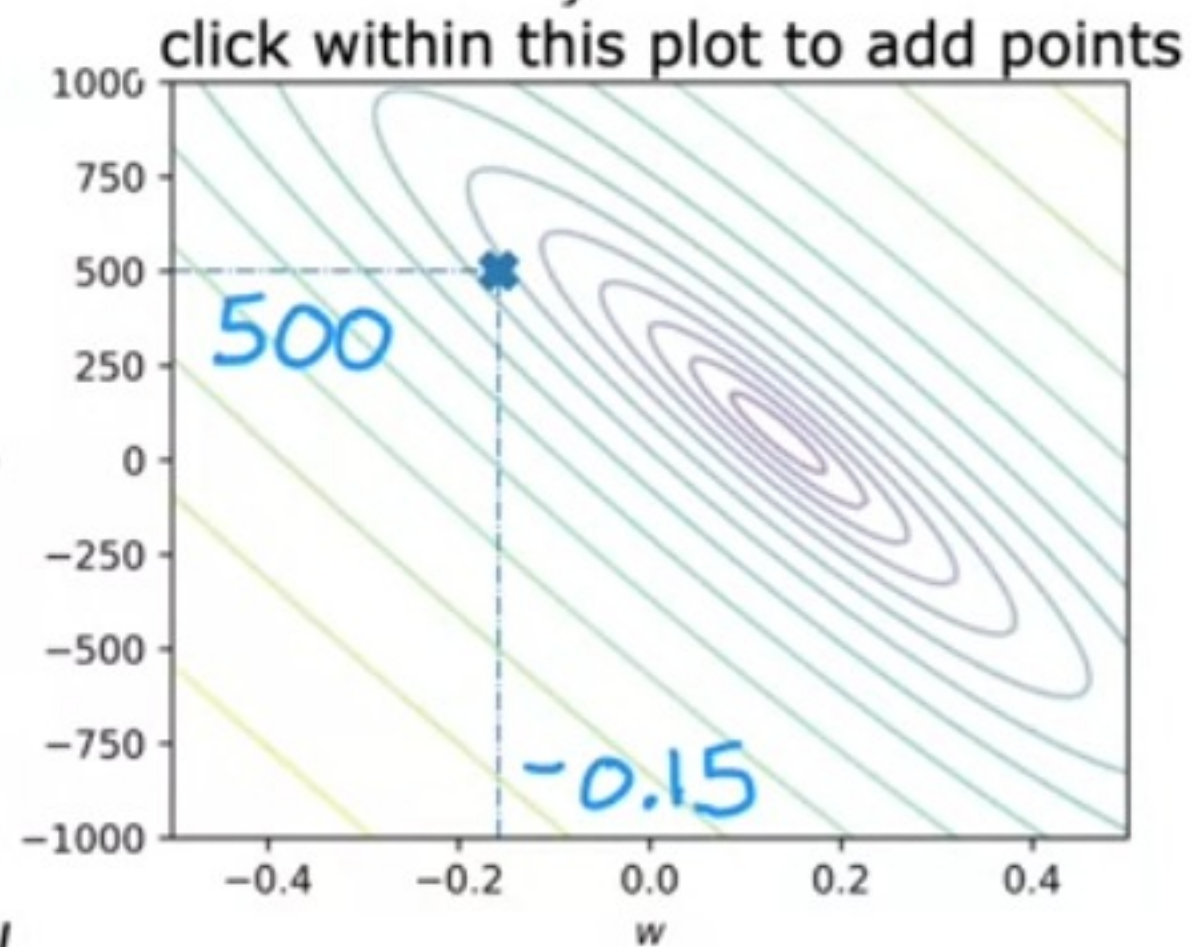
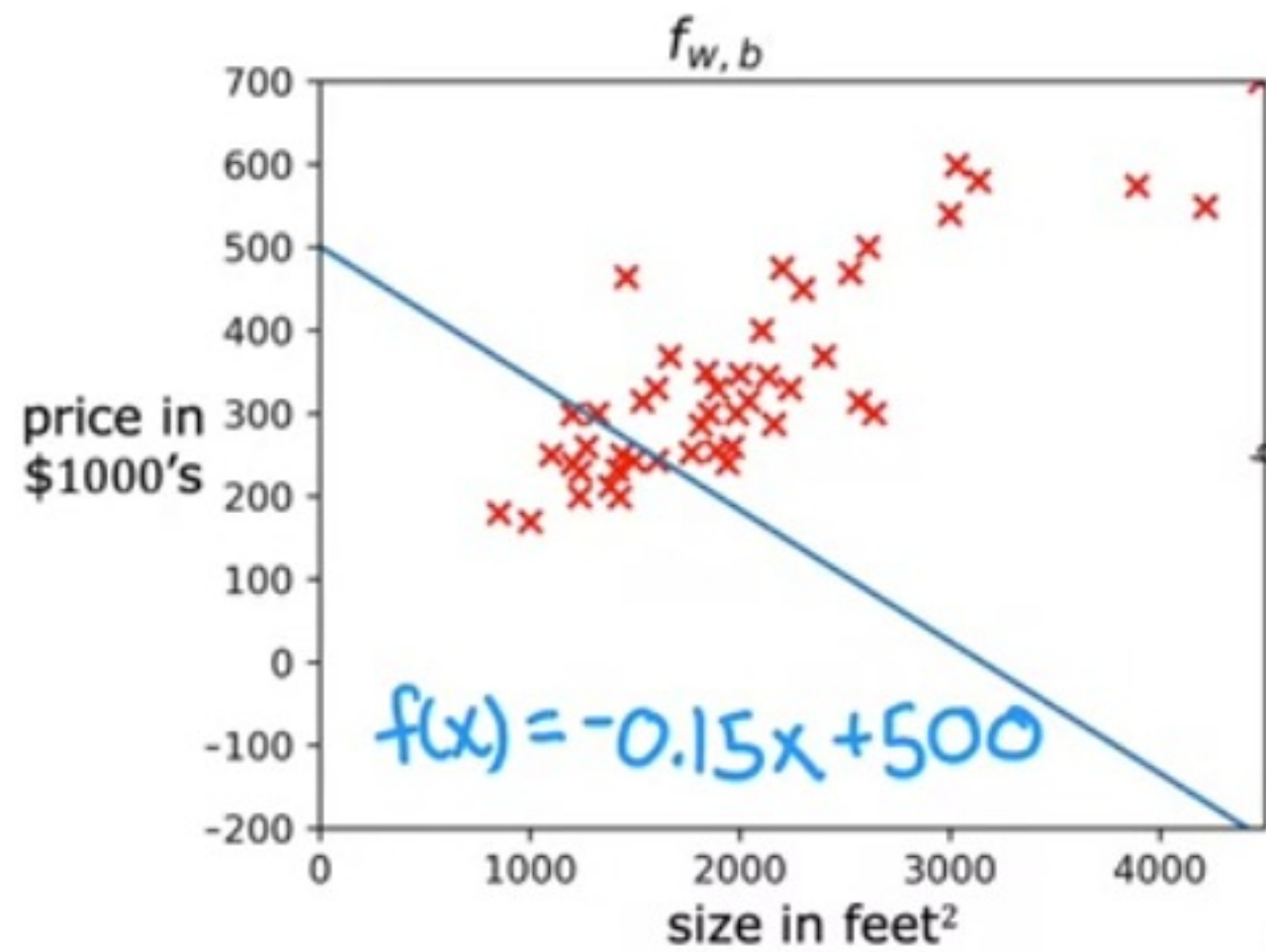
you can rotate this figure



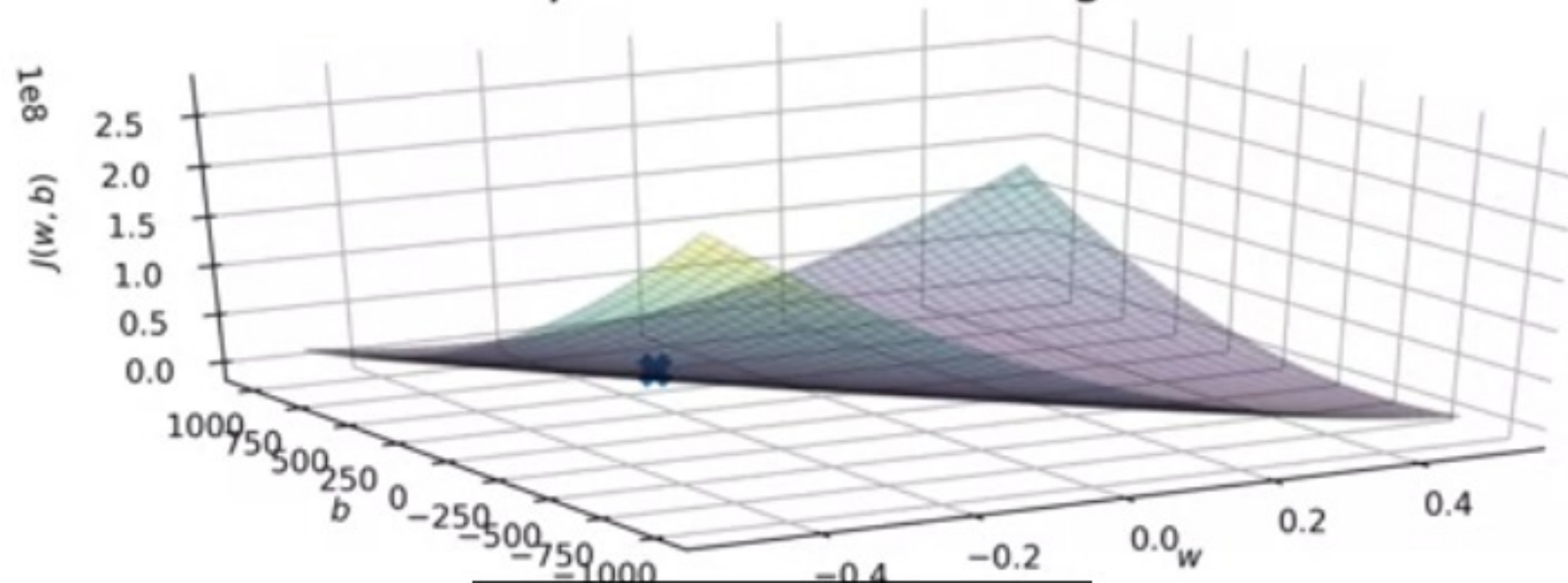


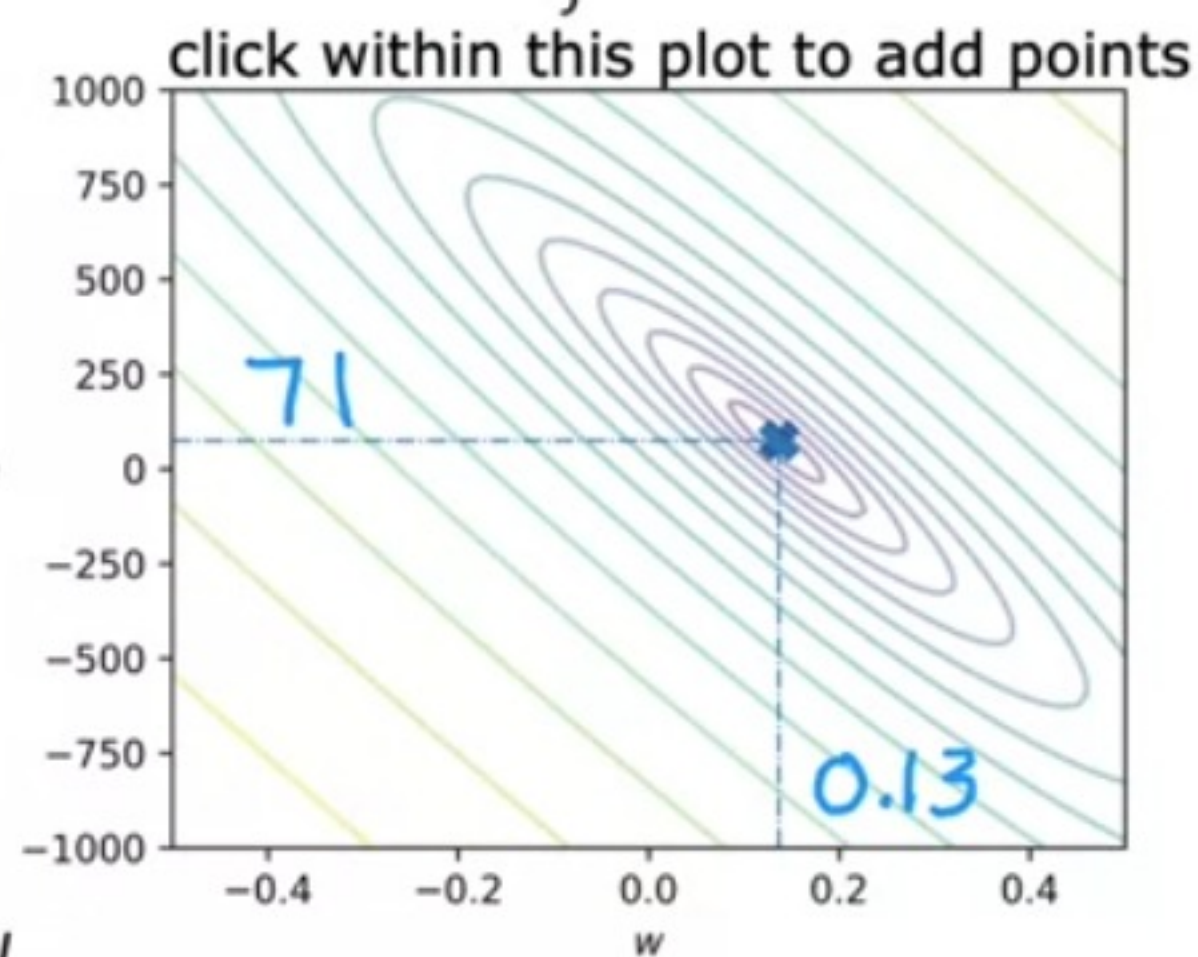
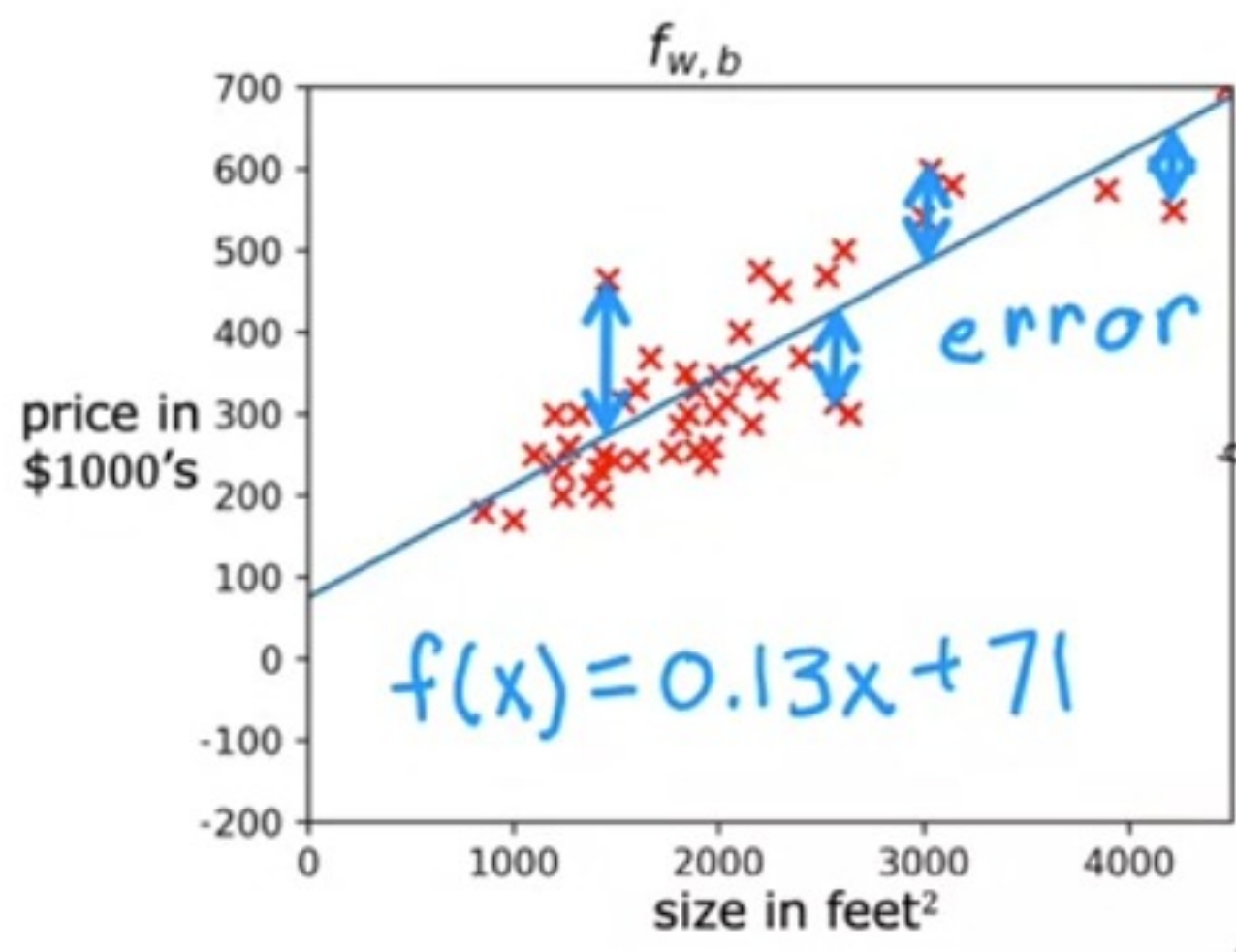
you can rotate this figure



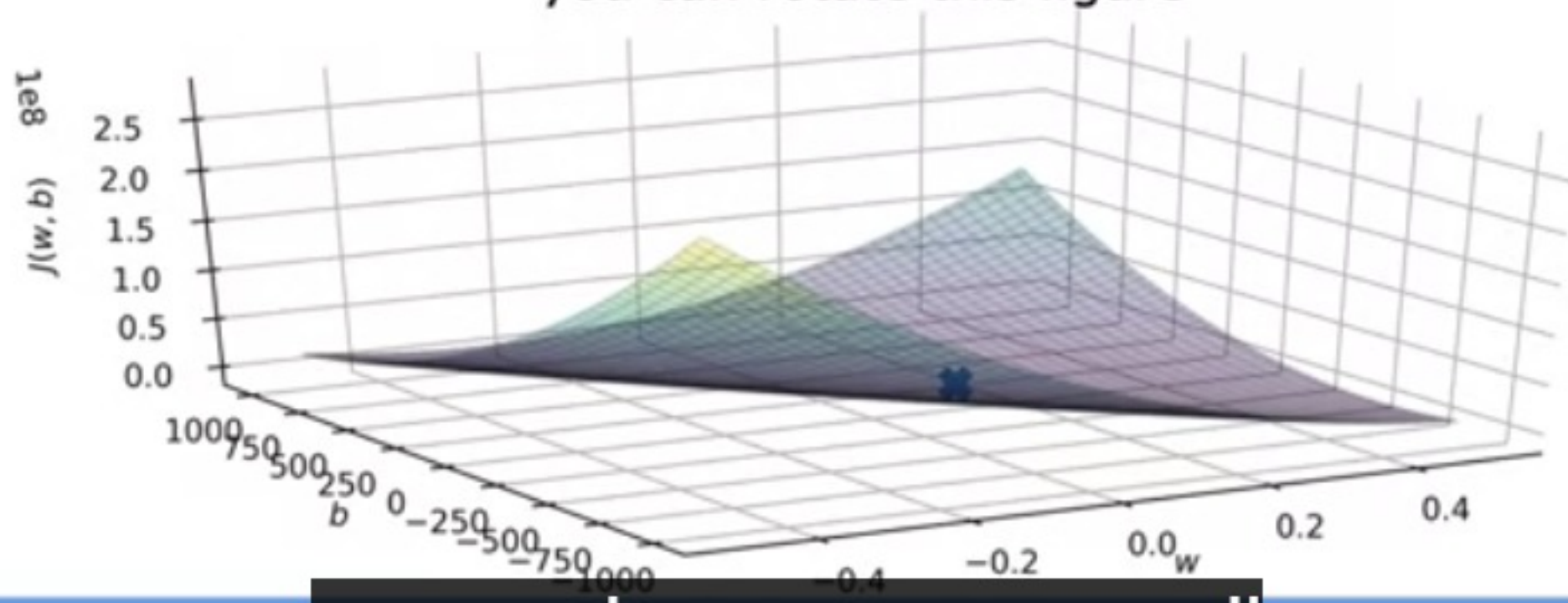


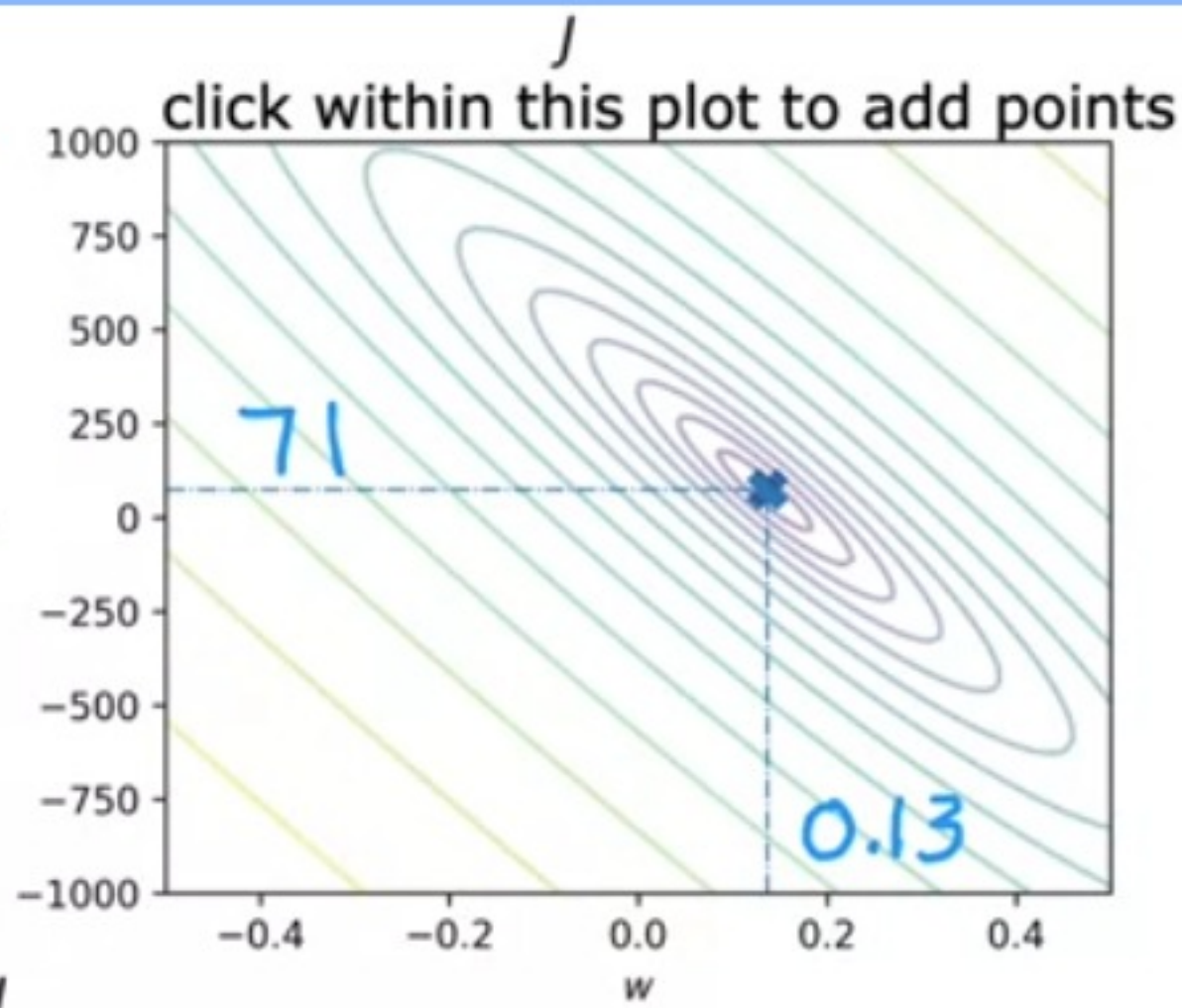
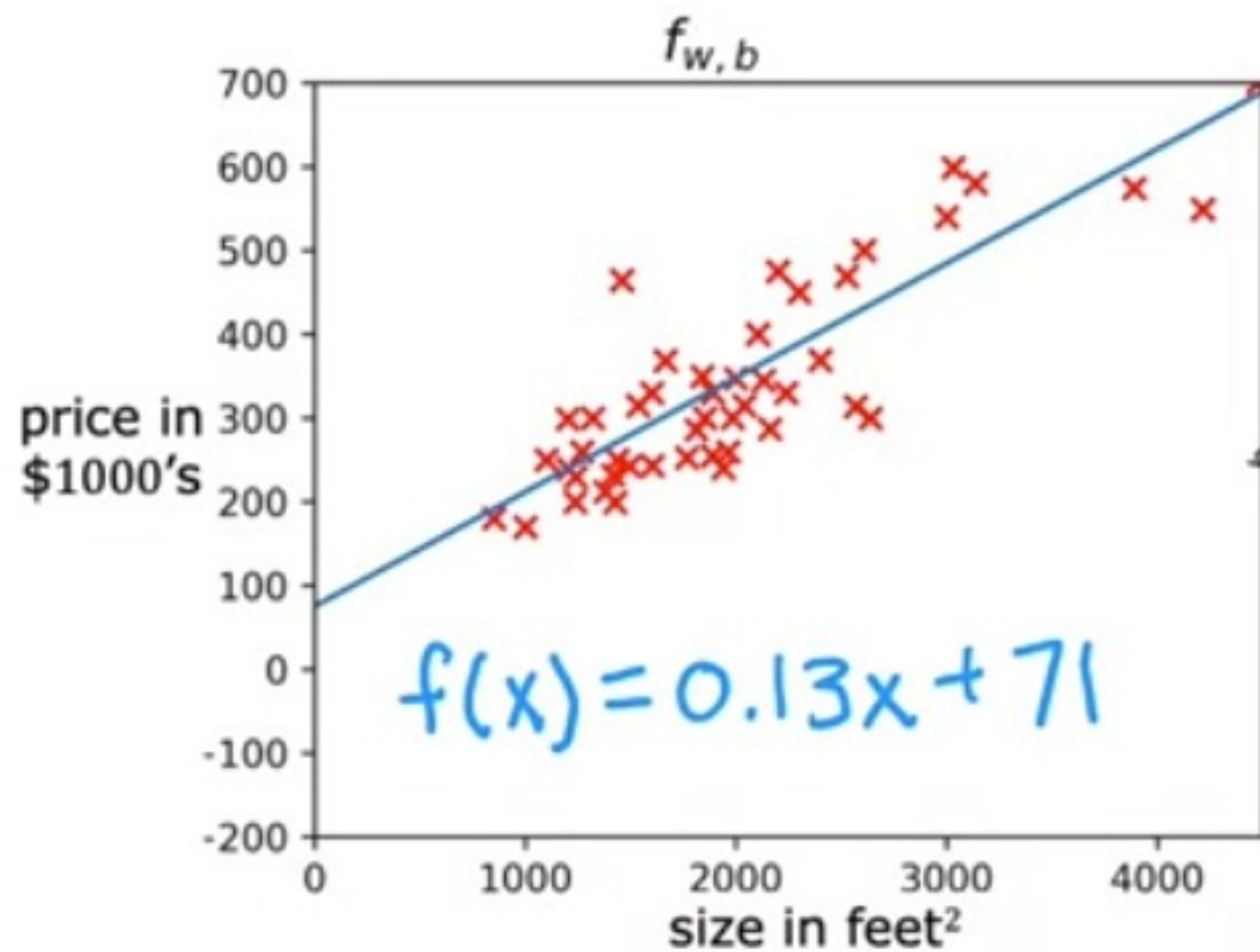
you can rotate this figure



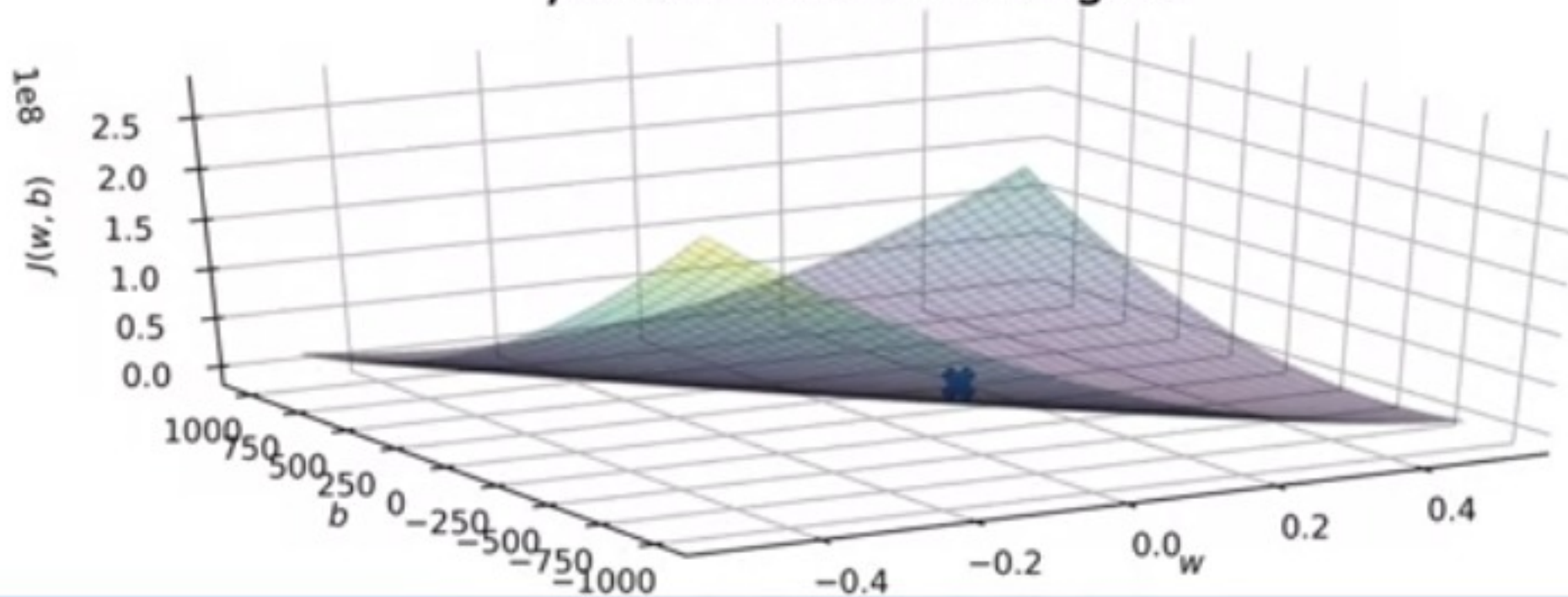


you can rotate this figure





you can rotate this figure



As we saw 3 more examples, as you can see the last one with

$$w = 0.13 \quad \& \quad b = 71$$

was actually a good fit for these data points

→ In this the sum of squared errors for all these data points is pretty close to the minimum possible sum of squared error among all possible straight line fits.

↳ as you can see on contour & 3D one it's close to centre & low heights.